Unpack Submissions: User Manual for Linux

Unpack Submissions turns the gradebook ZIP files you download from Blackboard into tidy folders: one per assignment, with each student's work named after their username. Source code is kept under its original name in a folder per student, so you can compile and run it as submitted. Documents, images, audio and video get PDF, JPEG, MP3 or MP4 copies you can open anywhere (plain-text files such as `.txt`, `.md` and `.csv` are kept as they are), and compiled files (`.class`, `.o` and the like) are thrown away.

It works on Debian, Ubuntu and their relatives, and on Fedora, Red Hat Enterprise Linux, Rocky and Alma Linux. You don't need to install anything first. The script checks what it needs and helps you get it.

Two extra programs

Most files are converted by the script itself, but two kinds need a separate program:

Files	Program needed for the PDF, MP3 or MP4 copy
Word, Excel and PowerPoint (`.docx`, `.xlsx`, `.pptx` and older `.doc`, `.xls`, `.ppt`), OpenDocument, RTF, Pages, Numbers and Keynote	LibreOffice
Audio and video (`.wav`, `.m4a`, `.mov`, `.webm` and the like)	ffmpeg

You don't need to install either in advance. The first time a batch contains such files, the script offers to download the program it needs (see below).

What you get

Put these files together in one folder:

- `unpack-submissions.sh` (the file you run)
- `unpack-submissions.py`

Unpacking a batch of submissions

1. Download the assignment ZIP files from Blackboard and move them into the folder with the scripts.
2. Open a terminal in that folder. In most file managers, right-click inside the folder and choose "Open in Terminal".

3. Type the following and press Enter:

```
bash unpack-submissions.sh gradebook_*.zip -o Grading
```

4. Answer any questions (see below).
5. When it has finished, open the `Grading` folder.

Questions it may ask

- **Python is missing**, or (on Debian and Ubuntu) Python's `venv` module is missing. The script shows the commands to install it and offers to run them for you with `sudo`. It asks for your password, and that is the only time it needs one.
- **ffmpeg is missing.** This is asked only when a batch contains audio or video. Type `Y` to download it (about 150 MB). It goes into a `Tools` folder beside the scripts and needs no password. If Fedora's own ffmpeg (`ffmpeg-free`) is already installed, the script uses it instead, with a different video encoder.
- **LibreOffice is missing.** This is asked only when a batch contains Word, Excel, PowerPoint or similar files, which can't be converted without it. Type `Y` to download it (about 350 MB) into the same `Tools` folder. This is available for x86-64 computers only; on ARM computers, install LibreOffice with the commands the script prints.

If you answer `N`, the script prints the commands for installing that program with your package manager (`apt` or `dnf`), plus a way to do it without a password, and carries on. Those files are still saved, but without a PDF or MP3/MP4 copy: Word, Excel and PowerPoint files in particular stay as originals only.

What is in each assignment folder

Item	What it is
`jdoe` (a folder)	jdoe's source code, under its original file names
`jdoe.docx`, `jdoe.mov`	jdoe's other files, renamed after the username
`jdoe-00.docx`, `jdoe-01.docx`	several files of the same type, in the order submitted
`jdoe.pdf`, `jdoe.mp4`	converted copies, for reading or playing; called `jdoe-docx.pdf` and so on only when two files would otherwise share a name
`Receipts`	Blackboard's receipt for each attempt
`manifest.csv`	every original file name, what it became and what was removed

Options

Option	What it does
`-o FOLDER`	where to put the assignment folders (normally the current folder)
`--force`	replace assignment folders left over from an earlier run
`--download yes`	download any missing converter without asking
`--download no`	never download; just give instructions
`--instructions`	show how to install ffmpeg and LibreOffice by hand
`--remove-converters`	delete everything the scripts downloaded (see below)
`--help`	list every option

When you have finished

When you've unpacked all your submissions, free up the space the converters use (several hundred megabytes):

```
bash unpack-submissions.sh --remove-converters
```

This deletes the `Tools` folder and the script's private copy of the Python libraries. It doesn't touch anything installed with `apt` or `dnf`, or the folders of submissions you've made. Anything needed later is downloaded again.

If something goes wrong

- **"already exists"**: an earlier run left that assignment folder. Add `--force` to replace it, or choose a different `-o` folder.
- **A download fails with a certificate error**: your network may inspect secure connections. Ask IT for the network's certificate file and run `export SSL_CERT_FILE=/path/to/that/file` before the script.
- **A download fails otherwise**: check your internet connection and run the script again, or use the instructions the script prints.

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